

K3 Compute cluster Event controller (CLEC)

ARCHITECTURE SPECIFICATION

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SECTION I: FUNCTIONAL OVERVIEW

1. Overview and Feature List

1.1 Reference Documents

Document	Link	Description
K3 INTD Specification	https://pds.design.ti.com/cgi-bin/internaldocs?docconfigid=694&filtertitle=2278	INTD spec
ARM GIC-400 r0p0 (DDI 0471A ID062111)	https://sps16.itg.ti.com/sites/CI/ARM/ARM%20specifications/Generic%20Interrupt%20Controller(GIC)/GIC400_Technical_Reference_Manual.pdf	GIC-400 spec
C7xV CPU specifications	https://pds.design.ti.com/cgi-bin/viewdocs?pds_bgid=21&docconfigid=694&filtertitle=2637&filteritem=15205	Delta Spec for updates in C7xV CPU
K3 Compute cluster specifications	https://pds.design.ti.com/cgi-bin/internaldocs?docconfigid=689&filtertitle=2208	Compute cluster spec
ARM v8 TRM (restricted access)	https://sps16.itg.ti.com/sites/CI/ARM/Forms/AllItems.aspx?RootFolder=%2fsites%2fCI%2fARM%2fARM%20specifications&FolderCTID=0x0120000368340F05F2C041B75237B47AC8AEAA&View=%7bB057050C-ED13-452E-882D-EF783EC1C18E%7d	ARM SharePoint
K3-W1 Soc interrupt specifications	https://pds.design.ti.com/cgi-bin/internaldocs?docconfigid=699&filtertitle=2225	

Table 1: CLEC Reference Documents

2. K3 Compute Cluster Event controller (CLEC):

2.1 Overview

For K3 Soc's that integrate an ARM core (A53/A72 etc), GIC will serve as the interrupt controller at the Soc level. GIC is integrated inside MSMC wrap and is part of compute cluster.

For K3 Soc's that integrate a C7x core, a low latency Cluster level interrupt controller (CLEC) is also integrated at MSMC wrap level or C7x wrap level depending on SOC definition. A high-level block diagram of CLEC is shown in **Error! Reference source not found.**:

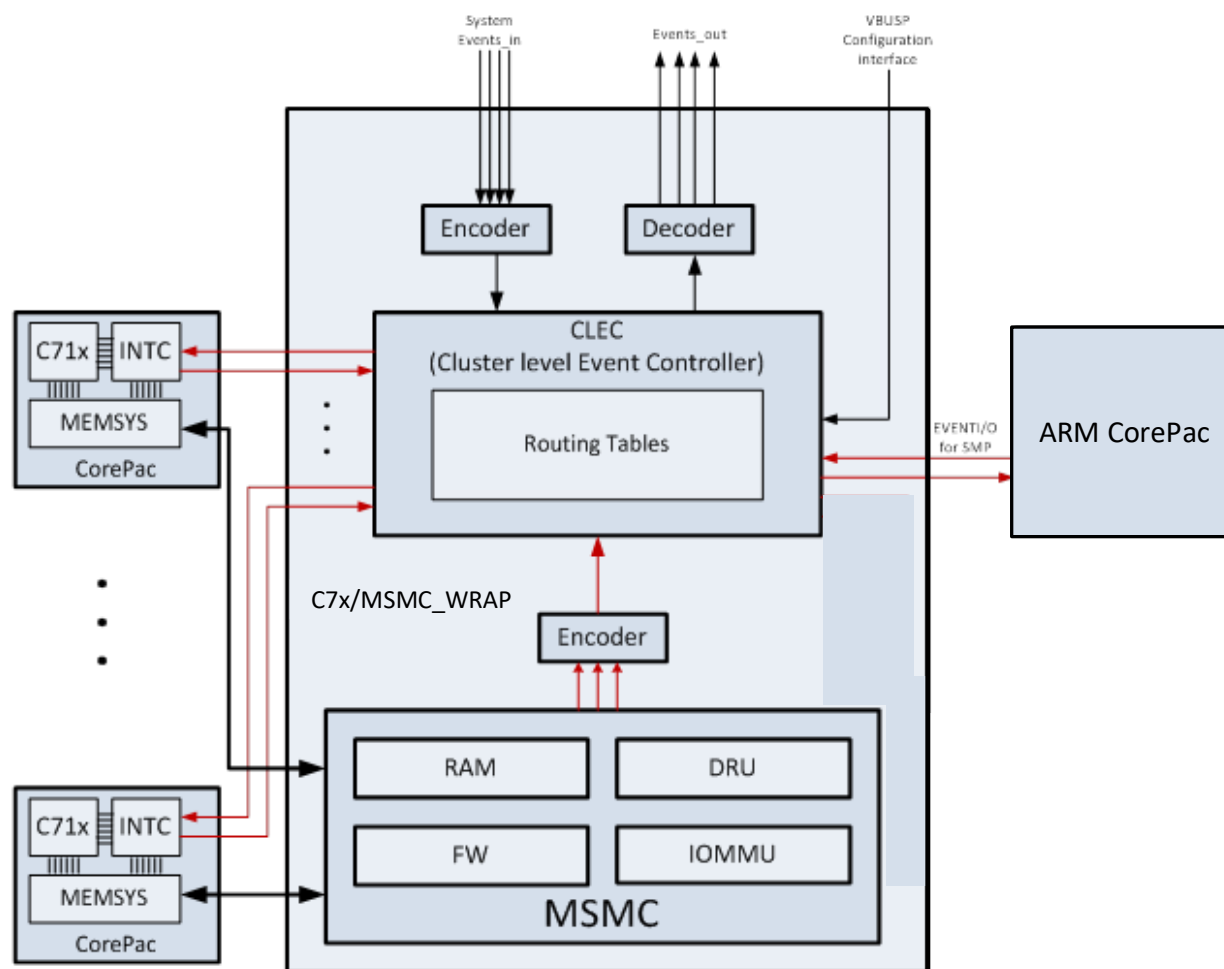


Figure 1 CLEC block diagram

The cluster event controller (CLEC) is responsible for:

- Software scalable event routing mechanism inside the compute cluster
- Aggregating incoming interrupt events from:
 - the system
 - MSMC (If present)
 - DRU
 - C71x Corepac IPs
- Distributing each aggregated interrupt event to one or more of the following:
 - the system (could be routed to the GIC externally)
 - DRU
 - ARM
 - C71x Corepac
- Originating interrupt events through the following mechanisms:
 - MMR writes to trigger new events
 - Reporting errors detected by the cluster event controller

2.2.1 CLEC configuration for K3-W1 Soc's:

The DSP interrupt controller CLEC supports the following features:

- 64 output events
 - Only level events exported out to Soc.
- Depending on the number of C7x cores integrated in the Soc, the number of events_in would be configurable in the compute cluster.
 - Number of input events configurable between 128, 256, 512, 1024, 2048 events.
 - K3 IPGeneric is used to capture the event – The IP Generic captures rising edge of the event. This ensures that there is no minimum pulse width requirement if it is a pulse event.

AM62A event mapping:

- CLEC can handle both can handle level/pulse input interrupts.
 - For pulse input interrupts, it is required that mapping is one to one and not many to one.
 - SOC out interrupt is always level regardless of input interrupt type (pulse or level).
 - Events routed to C7xV from CLEC are always pulse type
- Soc SPI input events are routed to both CLEC and GIC
- Exact mapping provided in Section 3.9.

	# events
CC internal events (C7x IPC, DRU events etc)	224
Soc input events	256
Optional, DSP only input events from Soc	32

TDA54 event mapping:

- SOC input and output events are increased to 128 from 64
- Total number of events are 2048
- CLEC can handle both can handle level/pulse input interrupts.
 - For pulse input interrupts, it is required that mapping is one to one and not many to one.
 - SOC out interrupt is always level regardless of input interrupt type (pulse or level).
 - Events routed to C7xV from CLEC are always pulse type
- Soc SPI input events are routed to both CLEC and GIC
- Exact mapping provided in Section 3.9.

3. Cluster level Event Controller (CLEC) Architecture

The cluster event controller routes events according to a RAM-based interrupt event routing table. This table holds one entry for each supported external interrupt/event number, plus some additional entries for CLEC internal events. External in this context means “external to the cluster event controller,” and corresponds to all interrupt events arriving at the cluster event controller from other sources. Internal refers to CLEC internal events that the event controller generates internally, or in response to an MMR write that triggers the event.

The routing table memory is sized at design time. The cluster event controller supports up to a maximum of 2K entries in the routing table from all interrupt event sources.

3.1 CLEC Block Diagram

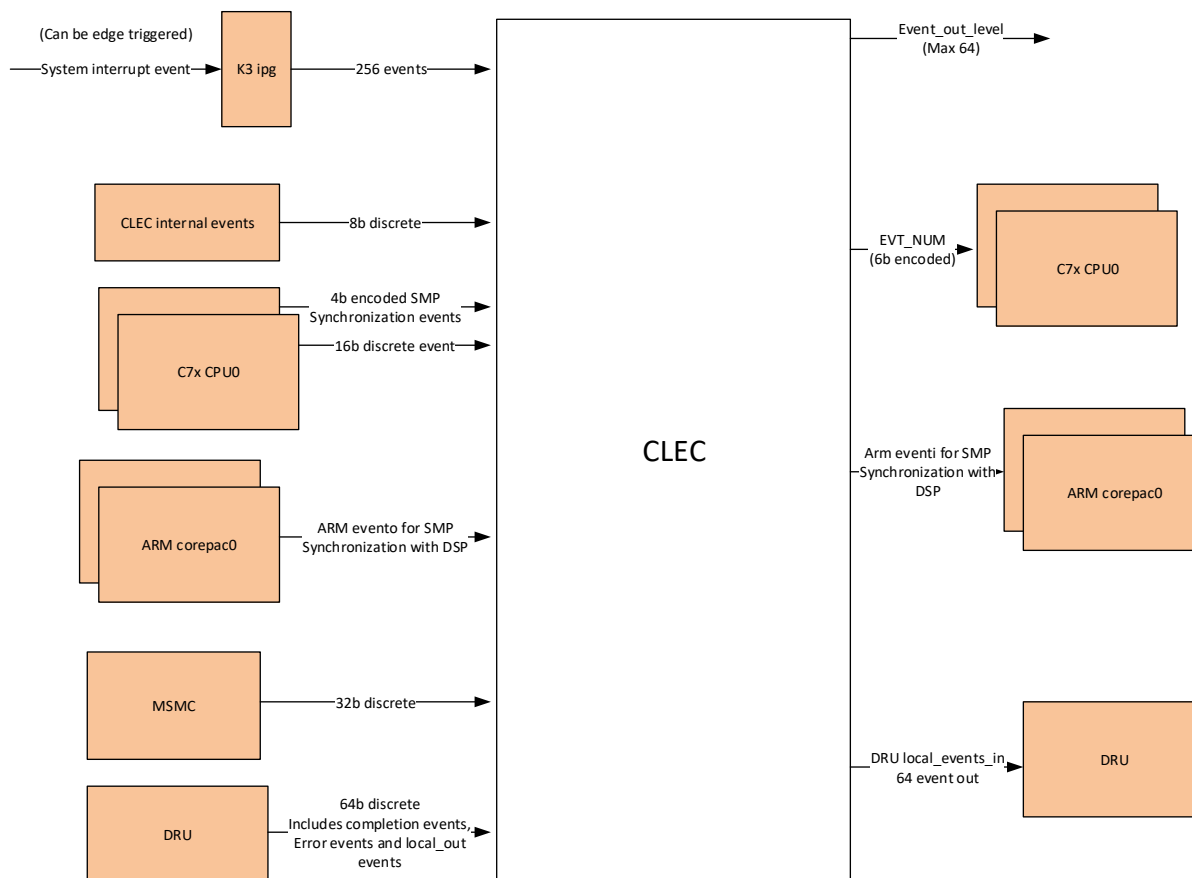


Figure 2 CLEC overview

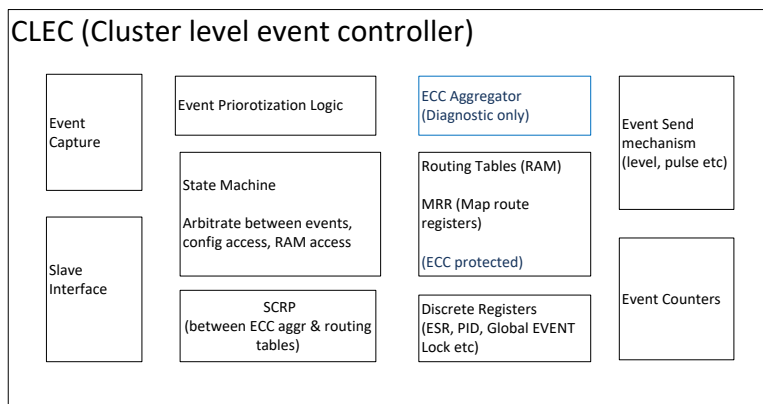


Figure 3 CLEC functional diagram

Figure 4 Compute Cluster Event Architecture

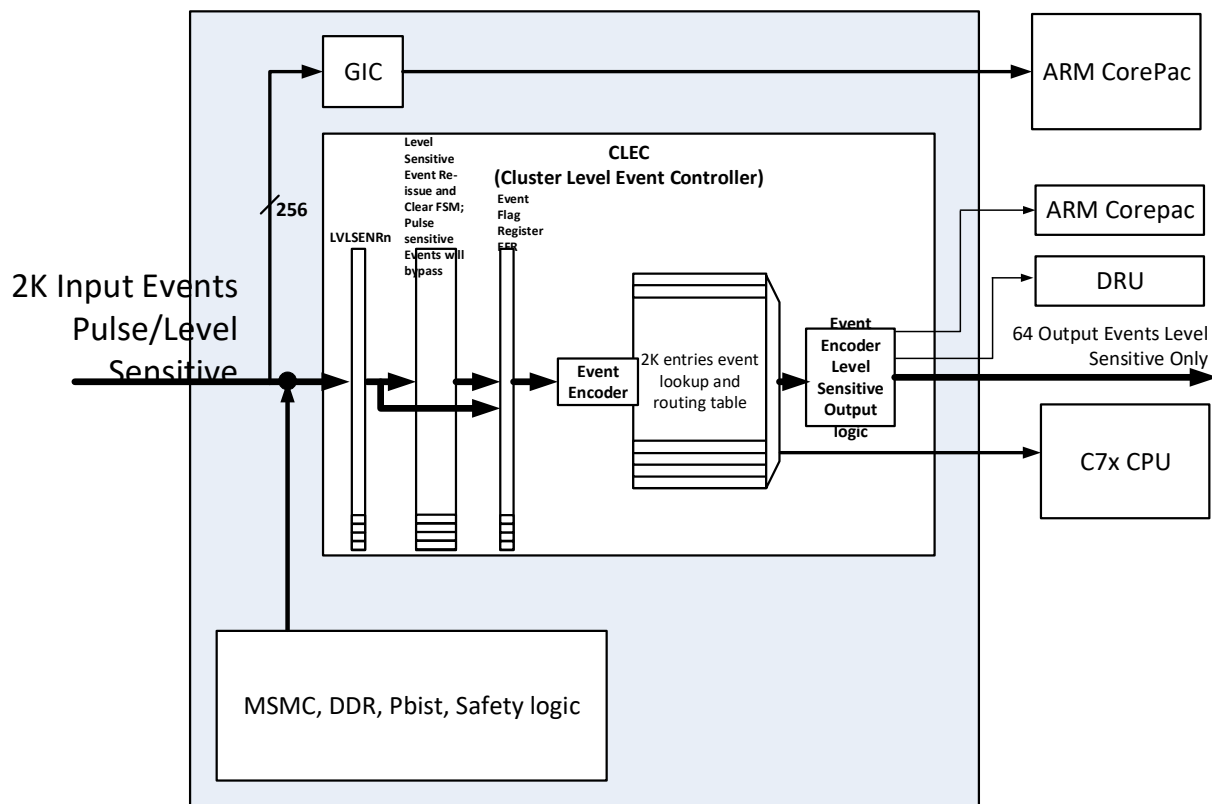
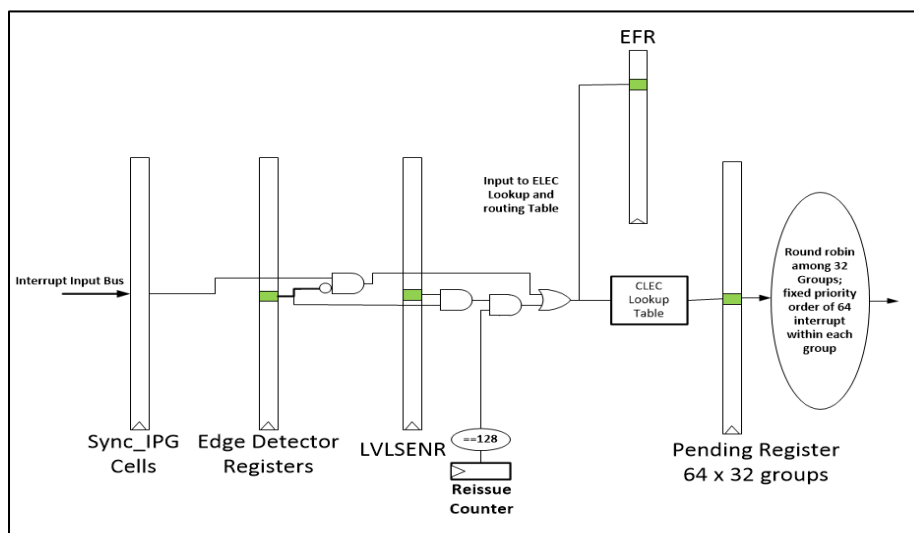


Figure 5 Level sensitive interrupt re-issue logic



Note: For level sensitive interrupts is re-issued every 128 cycles. This is to make sure level sensitive interrupts are not dropped by C7x. CLEC events to C7x are pulse events and for level sensitive events routed to C7x, event is sent every 128 cycles until interrupt from the source is de-asserted.

3.2 Interrupt Event Routing

Each routing table entry contains the following details:

- **Outgoing event number.** Each incoming event can be mapped to any outgoing event number.
- **Secure vs. not-secure.** Each incoming event can be claimed as a secure event.
 - Since C7xV doesn't support secure mode, by default all events are marked as non-secure
- **Destination port configuration.** This configuration allows for both single-cast and broadcast events.

The Cluster event Controller processes incoming events as follows. For each incoming interrupt event, the cluster event controller:

- Validates the interrupt event number. For out-of-range event numbers, the cluster event controller triggers an "Invalid Event Number" Event.
- Non-secure read/write to registers associated with a secure event will generate a Privilege error.
- Looks up the event in the routing table.
- For each port in the destination bitmap that has a bit set, it queues the event to that destination.
 - Queuing to multiple destinations happens in parallel, not sequentially, to minimize latency.
 - If no bits are set, the event is dropped silently, but an Interrupt dropped event is asserted. This event gets routed as determined by the routing table.

Since the routing table is a shared resource, the Cluster Event Controller only routes one new event per cycle. The routing state machine arbitrates among all incoming event sources round-robin, with the exception that internally generated error events have priority over all other event sources.

The Cluster Event Controller routes events to their destinations and also maintain pending interrupt information in Event Flag Register. Firmware needs to clear this pending interrupt bit in ISR.

3.3 CLEC – Programmer's view:

Here is a high-level programmer's view of the CLEC architecture:

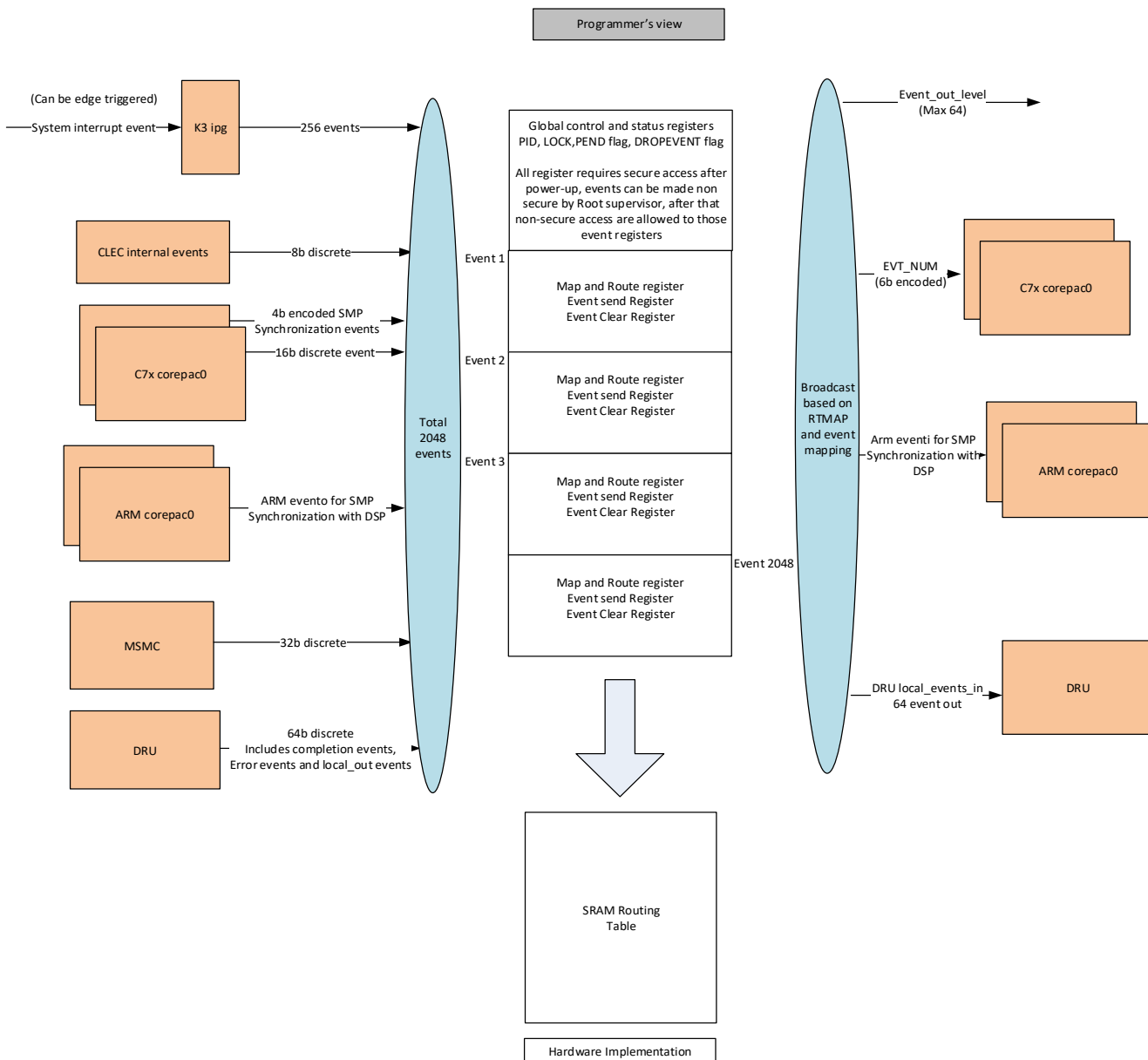


Figure 6 High level programmer's view of CLEC

3.4 CLEC configuration Interface

The Cluster Event Controller provides a simple memory mapped register interface over VBUSP_CFG interface (main config interface for K3W1 Soc). The CLEC configuration is a dedicated region select in the compute cluster vbusp configuration interface.

- This VBUSP interface is mapped to internal UMC memory region, CLEC access Path is through UMC.

vbusp_cfg base address	Name	AM62A, J7AEN, AM275	TDA54
Tie-off based (Look at Cluster spec for Base address)	Cluster level Event Controller (CLEC) – 1MB	Valid	Valid

3.5 Functional Safety and Error Injection

Standard K3 ECCAggregator/Wrapper strategy is used to implement ECC protection on the SRAM used to implement the routing table.

Note: ECC aggregator registers are part of the Global register list.

NOTE: Sam and Jay have confirmed that no need for parity/edc protection on the CLEC-c7x interface. This is due to the fact that none of the other encoded interfaces are protected. If a follow-on device is built with higher safety requirement, this is one of the safety improvements that could be added. On parity error, c7x_cpu could return machine fault.

3.6 Memory Protection

The Cluster Event Controller implements minimal memory protection checks. It relies on protection mechanisms outside to control accesses from various masters. Chip level firewalls control which masters may access the Cluster event Controller. MMUs provide finer-grain control for programmable processors that access the Cluster event Controller.

Specifically, the Cluster Event Controller returns the following errors:

- Address error on a read or write to unimplemented space
- Address error on a read to a write-only register
- Address error on a write to a read-only register
- Privilege error on a non-secure read or write to registers associated with a secure event (MRR.S = 1 for that event)
- Privilege error on a write to ESR on an event whose MRR.ESE = 0
- Privilege error on a write to MRR on an event when GELR.LOCK = 1

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Any further access control must be implemented with firewalls and/or MMU configuration outside the Cluster event Controller.

3.8 Cluster level event mapping table – CLEC output events:

Events out of the CLEC can be routed to Soc, DRU or C7x/ARM corepacs. The Soc event_out could be connected to the GIC or R5 depending on the criticality level of the event exported out of the cluster.

Internal to the compute cluster, the event_out bus is an encoded bus and it is decompressed at the cluster boundary to present one-hot 64bit or 128bit event_out bus to Soc. CLEC will export level outputs for soc_events_out bus.

For Soc, DRU, ARM events, the exact event number mapping is controlled by the MRR.EXT_EVNUM field as described in Section 4.3.1. MRR.EXT_EVNUM is an 8-bit field allowing 256 possible output events.

For C7x events, the exact event number mapping is controlled by the MRR.C7x_EVNUM field as described in Section 4.3.1. MRR.C7x_EVNUM is a 6-bit field allowing 64 possible output events.

Following is the mapping for S Maxwell and J7ES Soc's:

Table 2 CLEC Output Event mapping table for AM62A

category	MRR.EXT_EVNUM	# events	Event triggered	AM62A
Soc event_out	0-63	64	Soc output events (64 events)	Valid
DRU local event in	64-95	32	DRU local in trigger events	Valid
Reserved	96-127	32	DRU reserved	Reserved
ARM Corepac events - Corepac 0	128	1	ARM Corepac eventi for DSP	Valid
ARM Corepac events - Corepac 1	129	1	ARM Corepac eventi for DSP	Reserved
ARM Corepac events - Corepac 2	130	1	ARM Corepac eventi for DSP	Reserved
ARM Corepac events - Corepac 3	131	1	ARM Corepac eventi for DSP	Reserved
ARM Corepac events - Corepac 4	132	1	ARM Corepac eventi for DSP	Reserved
ARM Corepac events - Corepac 5	133	1	ARM Corepac eventi for DSP	Reserved
ARM Corepac events - Corepac 6	134	1	ARM Corepac eventi for DSP	Reserved
ARM Corepac events - Corepac 7	135	1	ARM Corepac eventi for DSP	Reserved
Reserved	255-136	120	Reserved	Reserved

Table 3 CLEC Output Event mapping table for TDA54

category	MRR.AC_DRU	MRR.EXT_E VNUM	# events	TDA54
Soc event_out	0	0-127	128	Valid
Central DRU local event in	0	128-159	32	Valid
Reserved	0	160-191	32	Valid
ARM Corepac events - Corepac 0	0	192	1	Valid
ARM Corepac events - Corepac 1	0	193	1	Reserved
ARM Corepac events - Corepac 2	0	194	1	Reserved
ARM Corepac events - Corepac 3	0	195	1	Reserved
ARM Corepac events - Corepac 4	0	196	1	Reserved
ARM Corepac events - Corepac 5	0	197	1	Reserved
ARM Corepac events - Corepac 6	0	198	1	Reserved
ARM Corepac events - Corepac 7	0	199	1	Reserved
Reserved	0	200-231	32	Reserved
Reserved	0	255-232	24	Reserved
Analytics Wrap0 node DRU	1	0-31	32	Valid
Analytics Wrap1 node DRU	1	32-63	32	Valid
Analytics Wrap2 node DRU	1	64-95	32	Valid
Analytics Wrap3 node DRU	1	96-127	32	Valid
Reserved	1	128-255	128	Reserved

3.8.1 Routing of Firewall exception events

Firewall exceptions are not routed through CLEC. Instead, they are routed from respective firewall slaves directly to DMSC. This is defined in the Compute cluster specification document referenced in Table-1.

3.8.1 Routing of Compute cluster critical events

K3 Safety requirements require the following:

- All critical events (1b/2b errors etc) be routed to ESM.
- CLEC can provide one level of aggregation, but it mandates that CLEC has to go through the exercise of proving that CLEC doesn't add any latent faults into our fault reporting mechanism.
- If DSP has to service any of the error events, then would be routed to CLEC through Soc_event_in.

Hence, error events are routed out of the compute cluster directly.

3.9 Cluster level event mapping table – Input events to CLEC

Each incoming interrupt event port is assigned a distinct range of event numbers at design time. This segregates interrupt events from multiple sources into different event number ranges. The exact size and offset of each range are determined and configured at design time.

3.9.1 AM62A input event mapping

CLEC software event number Software configurable in AM62A Any of events below can be routed to 64 events out of CC									
Evt num (from)	Evt num (to)	# events	Mapped to:	Validity in AM62A				Soc event out number	soc_events_out_level_intr
0	0	1	CLEC Reserved					0	SW Programmable
1	1	1	CLEC_ADDR_ERR	Valid				1	SW Programmable
2	2	1	CLEC_PRIV_ERR	Valid				2	SW Programmable
3	3	1	CLEC_INVALID_EVT	Valid				3	SW Programmable
4	4	1	CLEC_DROPPED_EVT	Valid				4	SW Programmable
5	5	1	dru_prot_err_intr	Valid				5	SW Programmable
6	6	1	Reserved					6	SW Programmable
7	7	1	Reserved					7	SW Programmable
8	8	1	arm_0_evento_in_intr	Valid				8	SW Programmable
31	9	23	Reserved	Reserved				9	SW Programmable
63	32	32	c7xv_corepac_events_in	Valid	-->	CLEC Router (SW programmable)	-->	10	SW Programmable
95	64	32	dru_complete_event_in	Valid				11	SW Programmable
127	96	32	dru_error_event_in	Valid				12	SW Programmable
159	128	32	dru_local_event	Valid				13	SW Programmable
191	160	32	Reserved	Reserved				14	SW Programmable
223	192	32	soc_events_in	Valid				15	SW Programmable
255	224	32	Reserved	Reserved				16	SW Programmable
511	256	256	gic_spi_intr	Valid				17	SW Programmable

Figure 8 Coarse event mapping for AM62A device

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3.9.2 TDA54 input event mapping

CLEC software event number Mapping is one to one in Maxwell Software configurable in J7 Any of events below can be routed to 64 events out of CC									
Evt num (from)	Evt num (to)	# events	Mapped to:	Validity in TDA54				Soc event out number	soc_events_out_level_intr/pulse
0	0	1	CLEC Reserved					0	SW Programmable
1	1	1	CLEC_ADDR_ERR	Valid				1	SW Programmable
2	2	1	CLEC_PRIV_ERR	Valid				2	SW Programmable
3	3	1	CLEC_INVALID_EVT	Valid				3	SW Programmable
4	4	1	CLEC_DROPPED_EVT	Valid				4	SW Programmable
5	5	1	Reserved					5	SW Programmable
6	6	1	Reserved					6	SW Programmable
7	7	1	Reserved					7	SW Programmable
8	8	1	arm0_dft_pbist_cpu_error_intr	Reserved				8	SW Programmable
9	9	1	arm1_dft_pbist_cpu_error_intr	Reserved				9	SW Programmable
10	10	1	arm2_dft_pbist_cpu_error_intr	Reserved	-->	CLEC Router (SW programmable)	-->	10	SW Programmable
11	11	1	arm3_dft_pbist_cpu_error_intr	Reserved				11	SW Programmable
12	12	1	c7x_4_dft_pbist_cpu_error_intr	Valid				12	SW Programmable
13	13	1	c7x_5_dft_pbist_cpu_error_intr	Valid				13	SW Programmable
14	14	1	c7x_6_dft_pbist_cpu_error_intr	Valid				14	SW Programmable
15	15	1	msmc_dft_pbist_cpu_error_intr	Reserved				15	SW Programmable
16	16	1	msmc_ctset_intr	Reserved				16	SW Programmable
17	17	1	dru_c_protocol_error	Valid				17	SW Programmable
18	18	1	dru_node0_protocol_error	Valid				18	SW Programmable
19	19	1	dru_node1_protocol_error	Valid				19	SW Programmable
20	20	1	dru_node2_protocol_error	Valid				20	SW Programmable
21	21	1	dru_node3_protocol_error	Valid				21	SW Programmable
22	22	1	c7x_7_dft_pbist_cpu_error_intr	Valid				22	SW Programmable
23	23	1	Reserved					23	SW Programmable
24	24	1	msmc_null_slave_error	Reserved				24	SW Programmable
25	25	1	arm4_dft_pbist_cpu_error_intr	Reserved					
26	26	1	arm5_dft_pbist_cpu_error_intr	Reserved					
27	27	1	arm6_dft_pbist_cpu_error_intr	Reserved					
28	28	1	arm7_dft_pbist_cpu_error_intr	Reserved					
29	29	1	Reserved	Reserved				128-25	SW Programmable
55	30	26	msmc_event_in_intr	Reserved					
63	56	8	Reserved	Reserved					
95	64	32	dru_c_complete_event_in[31:0]	Valid					
127	96	32	Reserved	Reserved					
159	128	32	dru_c_error_event_in[31:0]	Valid					
191	160	32	Reserved	Reserved					
223	192	32	dru_c_localout_event_in[31:0]	Valid					
255	224	32	Reserved	Reserved					
287	256	32	C7x_0 - 32 discrete events (including IPC)[31:0]	Reserved					
319	288	32	C7x_1 - 32 discrete events (including IPC)[31:0]	Reserved					
351	320	32	C7x_2 - 32 discrete events (including IPC)[31:0]	Reserved					
383	352	32	C7x_3 - 32 discrete events (including IPC)[31:0]	Reserved					
415	384	32	C7x_4 - 32 discrete events (including IPC)[31:0]	Valid					
447	416	32	C7x_5 - 32 discrete events (including IPC)[31:0]	Valid					
479	448	32	C7x_6 - 32 discrete events (including IPC)[31:0]	Valid					
511	480	32	C7x_7 - 32 discrete events (including IPC)[31:0]	Valid					



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512	512	1	ARM 0 - event_i	Valid					
513	513	1	ARM 1 - event_i	Reserved					
514	514	1	ARM 2 - event_i	Reserved					
515	515	1	ARM 3 - event_i	Reserved					
516	516	1	ARM 4 - event_i	Reserved					
517	517	1	ARM 5 - event_i	Reserved					
518	518	1	ARM 6 - event_i	Reserved					
519	519	1	ARM 7 - event_i	Reserved					
520	520	1	msmc_pbist_cpu4_intr	Reserved					
521	521	1	msmc_pbist_cpu5_intr	Reserved					
522	522	1	msmc_pbist_cpu6_intr	Reserved					
523	523	1	msmc_pbist_cpu7_intr	Reserved					
524	524	1	arm_0_pbist_1_cpu_intr_sync	Reserved					
525	525	1	arm_1_pbist_1_cpu_intr_sync	Reserved					
531	526	6	Reserved	Reserved					
532	532	1	aw4_msmc1_ctm_intr_intr	Reserved					
533	533	1	aw5_msmc1_ctm_intr_intr	Reserved					
534	534	1	aw6_msmc1_ctm_intr_intr	Reserved					
535	535	1	aw7_msmc1_ctm_intr_intr	Reserved					
599	536	64	Soc DSP input events[63:0]	Valid					
631	600	32	dru_node0_complete_event_in[31:0]	Valid					
663	632	32	dru_node0_error_event_in[31:0]	Valid					
695	664	32	dru_node0_localout_event_in[31:0]	Valid					
727	696	32	dru_node1_complete_event_in[31:0]	Valid					
759	728	32	dru_node1_error_event_in[31:0]	Valid					
791	760	32	dru_node1_localout_event_in[31:0]	Valid					
823	792	32	dru_node2_complete_event_in[31:0]	Valid					
855	824	32	dru_node2_error_event_in[31:0]	Valid					
887	856	32	dru_node2_localout_event_in[31:0]	Valid					
919	888	32	dru_node3_complete_event_in[31:0]	Valid					
951	920	32	dru_node3_error_event_in[31:0]	Valid					
983	952	32	dru_node3_localout_event_in[31:0]	Valid					
1023	984	40	Reserved	Reserved					
1983	1024	960	Soc GIC SPI events[959:0]	Valid					
2047	1984	64	Soc DSP input events[127:64]	Valid					



3.9.2.1 C7xv corepac input event mapping table

C7x event mapping to CLEC	Event triggered	AM62A, J7AEN, TDA54
0	umc_err_intr[0]	Valid
1	umc_err_intr[1]	Valid
2	dmc_err_intr[0]	Valid
3	dmc_err_intr[1]	Valid
4	dmc_err_intr[2]	Valid
5	dbg_ctm_intr	Valid
6	dbg_aet_intr	Valid
7	Reserved	Reserved
8	Reserved	Reserved
9	Reserved	Reserved
10	Reserved	Reserved
11	Reserved	Reserved
12	Reserved	Reserved
13	Reserved	Reserved
14	Reserved	Reserved
15	Reserved	Reserved
31-16	IPC events	valid

3.10 ESM Event mapping

3.10.1 AM62A esm_event_out mapping

esm_events_out_level_intr	Mapped to:	esm_events_out_pulse_intr	Mapped to:	validity in AM62A
0	c7xv_eccaggr0_uncorrected_err_level_intr	0	c7xv_eccaggr0_uncorrected_err_pulse_intr	Valid
1	c7xv_eccaggr0_corrected_err_level_intr	1	c7xv_eccaggr0_corrected_err_pulse_intr	Valid
2	Reserved	2	Reserved	Reserved
3	Reserved	3	Reserved	Reserved
4	Reserved	4	Reserved	Reserved
5	Reserved	5	Reserved	Reserved
6	Reserved	6	Reserved	Reserved
7	Reserved	7	Reserved	Reserved

4. Memory Map

4.1 CLEC Memory Map

4.1.1 Global Registers

The Cluster Event Controller has a small number of global control registers.

Event-0 is not a valid event. If asserted will result in "Invalid event error".

The Global registers are mapped to a dedicated 64KB window. The intent here is to minimize the additional address space required for the global controls. In a system with virtualization, it's expected that the hypervisor always claims this window.

- PID, GELRS, GELRNS, EFR and EDR are the global control/status registers and are defined in Section 4.2.

Name	Description	Address	Size (Bytes)	Permissions	Notes
PID	CLEC PID register	0x0000	4	RO	Global register
EFR	CLEC EFR register	0xA000	256	RO	Global register
EDR	CLEC EDR register	0xB000	256	RO	Global register
GELRS	Global Event Lock register for secure claim	0xC000	4	RW	Global register
GELRNS	Global Event Lock register for non-secure claim	0xD000	4	RW	Global register

4.1.2 Event #1- #2047 - Per-Event Registers

Each event's 64KB window consists of the following registers:

(Note: Event registers are only word accessible)

Name	Description	Address	Size (Bytes)	Permissions	Notes
MRR _n	Map and Route Register	$0x80000 + 0x0100 * (n) + 0x0$	4	RW	See format below
ESR _n	Event Send Register	$0x80000 + 0x0100 * (n) + 0x4$	4	WO	Write-only command register
ECR _n	Event Clear Register (Clear level interrupts)	$0x80000 + 0x0100 * (n) + 0x8$	4	WO	Write-only command register

4.2 Global Registers

4.2.1 PID (CLEC PID register), (0x0000)

This is a read-only value that identifies the Cluster level event Controller. The Revision Register contains the major and minor revisions for CLEC.

Bits	Field	Type	Reset	Description
31:30	scheme	R	01	Always read as 01. Writes have no effect.
29:28	BU	R	10	Always read as 10 = "Processors"
27:16	func	R	0x010	Always read as the assigned func id.
15:11	rtl	R	0	RTL version of the module. The "R" value in the X.Y.R.Z as defined by K3.
10:8	major	R	1	Major revision of module. The "X" value in the X.Y.R.Z defined in K3.
7:6	custom	R	0	Special version.
5:0	minor	R	0	Minor revision of module. The "Y" value in the X.Y.R.Z defined in K3

4.2.2 EFR (CLEC Event Flag Register), (0xA000 – 0xA0FF)

This is a read-only register which contains pending flag for all events queued by CLEC. This flag is set regardless of corresponding event is enabled or disabled.

- ⇒ For pulse events – flag register is cleared as soon as event is sent
- ⇒ For level events – flag register is cleared when interrupt event is deasserted
- ⇒ Above is to support scenario where CPU wants to disable some events temporarily while doing some important task and then re-enable the event, however when CPU re-enables the event it doesn't want to miss any events occurred when it was disabled temporarily.
- ⇒ This flag can be cleared by writing 0x1 to corresponding ECR_n register before enabling the event; this will avoid any spurious event during power-up/power-down to reach to CPU.

This register supports vector access as well where system can read 512 bits at a time. This register is 2048 bits wide (256 bytes) where each bit corresponds to one of the 2k input events.

Bits	Field	Type	Reset	Description
511:2	EFR_event_n	R	0	Event n pending flag
1	EFR_event_1	R	0	Event 1 pending flag
0	EFR_event_0	R	0	Event 0 pending flag

To access EFR bit of event n, firmware needs to access address $\text{address} = (0xA000 + (\text{event_number}/32)*4)$ and bit number would be $(\text{event_number}\%32)$.

Note: flags for secure events can only be read by secure supervisor. Non-secure supervisor always gets value 0 for secure event flags.

4.2.3 EDR (CLEC Event Drop Register), (0xB000 – 0xB0FF)

This is a read-only register which contains dropped event flag for each event. For input events which are pulse and not level, if CLEC receives back to back pulses of event and second pulse comes before event flag for the first event is cleared (which means event is not sent from CLEC queue yet), there is no way to tell destination that there were two events requested. CLEC will generate dropped event interrupt in this case and also sets a corresponding event dropped flag in EDR register. This flag bit is set until system clears it by writing 0x2 to corresponding ECR_n register. This register supports vector access as well where system can read 256 bits at a time. This register is 512 bits wide (64 bytes) where each bit corresponds to one of the 512 input events.

Bits	Field	Type	Reset	Description
2047:2	EDR_event_n	R	0	Event n dropped event flag
1	EDR_event_1	R	0	Event 1 dropped event flag
0	EDR_event_0	R	0	Event 0 dropped event flag

To access EDR bit of event n, firmware needs to access address $\text{address} = (0xB000 + (\text{event_number}/32) * 4)$ and bit number would be $(\text{event_number} \% 32)$.

4.2.4 GELRS (Global Event Lock register for secure claims), (0xC000)

When set, all the event configurations that are marked with MRR.S=1 become read-only. Attempts to write the MRR register for that event will return a privilege error. When clear, MRR may be read or written. This is an advisory lock to avoid spurious writes.

The purpose of GELRS is to allow a hypervisor to expose an event to a virtual machine container, while preventing that virtual machine from changing the configuration. The hypervisor must provide a writeable page to the contained virtual machine so that the virtual machine can use the Event Send Register. The GELRS acts as a global control, removing write permission to MRR when MRR.S=1. Since only Root supervisor has access to the Global registers, only root supervisor may change the value of the GELRS.LOCK bit.

Bits	Field	Type	Reset	Description
31:1	RSVD	R	0	Reserved
0	LOCK	RW	0	GELRS.LOCK

4.2.1 GELRNS (Global Event Lock register for non-secure claims), (0xD000)

When set, all the event configurations that are marked with MRR.S=0 become read-only. Attempts to write the MRR register for that event will return a privilege error. When clear, MRR may be read or written. This is an advisory lock to avoid spurious writes.

The purpose of GELRNS is to allow a hypervisor to expose an event to a virtual machine container, while preventing that virtual machine from changing the configuration. The hypervisor must provide a writeable page to the contained virtual machine so that the virtual machine can use the Event Send Register. The GELRNS acts as a global control, removing write permission to MRR when MRR.S=0. Since only Root supervisor has access to the Global registers, only root supervisor may change the value of the GELRNS.LOCK bit.

Bits	Field	Type	Reset	Description
31:1	RSVD	R	0	Reserved
0	LOCK	RW	0	GELRNS.LOCK

4.3 Event control registers

4.3.1 MRR_n: Map and Route Register (0x80000 + 0x0100 * (n) + 0x0)

Bits	Field	Type	Reset	Description
31	S	R	0 (Default: non-secure)	Secure Claim. If set to 1, a non-secure read/write to MRR will generate a Privilege error Since AM62A CPU doesn't support secure mode, this bit is made reserved.
30	ESE	RW	0	Event-Send enable. Controls whether CLEC sends this event when ESR is written or Event occurred on interrupt line.
29:26	Reserved	R	0	Reserved
25	EPF	R	0	MRR.EPF - same value as EFR, read only pending status of all events regardless of they are enabled or disabled, for pulse events this flag is cleared as soon as event is sent, for level events this flag is cleared when input event is deasserted, writing 1 to ECR_n register also clears this bit in case of pending pulse event
24	Is_lvl	RW	1	Indicates if input event is level (1) or pulse (0). It is expected from firmware to initialize this information accurately for proper CLEC operation. level sensitive interrupts are re-issued every 128 cycles. This is to make sure level sensitive interrupts are not dropped by C7x. CLEC events to C7x are pulse events and for level sensitive events routed to C7x, event is sent every 128 cycles until interrupt from the source is de-asserted.
23	Reserved	R	0	Reserved
22	AC_DRU	RW	0	Selects Analytics wrap node DRUs when set
21:16	RTMAP	RW	0x01 (Default: Send to None)	If Bit [16] = 1 → Send to None If Bit [17] = 1 → Send to System, (only if Bit [16] = 0) If Bit [21:18] = Send to C7x Corepac # encoded value (only if Bit [16] = 0) = F → Broadcast to all C7x Corepacs. = For TDA5 C7x mapping is from 4 to 8
15:8	EXT_EVNUM	RW	n%128	Encoded external event number to send when this event is triggered (max=256) Mapping: Refer Section 3.8. (routing requires if Bit [16] = 0 and Bit [17] = 1) 0-127: SOC events out 128-159: DRU events 192: ARM event Rest are reserved
7:6	Reserved	R	0	Reserved
5:0	C7x_EVNUM	RW	0	C7x event number to send when this event is triggered (max=64)

The MRR is readable by any privilege level. It is writeable by supervisor mode if unlocked by GELRS or GELRNS registers defined in section 4.2.

Since MRR register accesses are mapped to SRAM, it only supports word accesses.

Note:

Default value for **Central DRU** events are different then reset of the events per firmware team request. These DRU events are enabled and routed to C7x by default. Because DRU events are internal to msmc_wrap, bring up of MSMC/CLEC and DRU is at the same time and synchronous, so there is no concern of glitch or spurious interrupt.

dru_complete_event num 64 to 80 → ESE = 1, Is_lvl = 0, RTMAP = 3C(broadcast to DSPs), C7x_EVNUM = n%64.

dru_error event num 96 to 112 → ESE = 1, Is_lvl = 0, RTMAP = 3C(broadcast to DSPs), C7x_EVNUM = n%64 + 16.



dru_local event num 128 to 144 → ESE = 1, Is_lvl = 0, RTMAP = 3C(broadcast to DSPs), C7x_EVNUM = n%64 + 32.
For events which are pulse, mapping should be one to one because aggregation of pulse event cannot happen as SOC output is always level.

4.3.2 ESR_n (Event send register), (0x80000 + 0x0100 * (n) + 0x4)

The ESR is a write-only register. It ignores the value written (stateless). When written, the Cluster Event Controller will

- Generate an internal event specified by MRR.C7x_EVNUM, if MRR.ESE is set for that event.
- Generate an external event specified by MRR.EXT_EVNUM, if MRR.ESE is set for that event

Otherwise, the Cluster Event Controller does not send an event in response to the write; instead, it returns a privilege error for the write status.

Note: Corepac powerup/power down sequence should comprehend CLEC sequencing to avoid interrupt dropped events.

Bit	Name	Type	Reset	Description
31:0	reserved	W	0x0	Reserved

4.3.3 ECR_n (Event clear register), (0x80000 + 0x100 * (n) + 0x8)

The ESR is a write-only register. When written value 0x1, the Cluster Event Controller will clear any level interrupt set for that event.

Bit	Name	Type	Reset	Description
31:2	reserved	W	0x0	Reserved
1	EDR_clear	W	0x0	W1 to this bit clears EDR bit
0	EFR_clear	W	0x0	W1 to this bit clears event as well as EFR bit

5. Programmer's Guide

5.1 Initialization sequence

DMSC or an external master can program the routing table before booting a core. The CLEC routing tables can be configured by any external master (or) C7x (or) ARM core.

Refer to Figure 6 for the software programmer view.

Typical software sequence is that the routing table is programmed a-priori – before start of any application or program.

5.2 Real-time operating requirements

5.3 Power up/down sequence

- CLEC is part of MSMC power domain. If MSMC is powered down, CLEC entries have to be reinitialized by software.